FUZZY EXPERT SYSTEM FOR STOCK PORTFOLIO SELECTION: AN APPLICATION TO BOMBAY STOCK EXCHANGE

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Selection of proper stocks, before allocating investment ratios, is always a crucial task for the investors. Presence of many influencing factors in stock performance have motivated researchers to adopt various Artificial Intelligence (AI) techniques to make this challenging task easier. In this paper a novel fuzzy expert system model is proposed to evaluate and rank the stocks under Bombay Stock Exchange (BSE). Fuzzy Delphi method is used to identify the critical factors initially and then factors with lower correlation coefficients are finally considered for further consideration. Dempster-Shafer (DS) evidence theory is used for the first time to automatically generate the consequents of the fuzzy rule base to reduce the effort in knowledge base development of the expert system. Later a portfolio optimization model is constructed where the objective function is considered as the ratio of the difference of fuzzy portfolio return and the risk free return to the weighted mean semi-variance of the assets that has been used. The model is solved by applying Ant Colony Optimization (ACO) algorithm by giving preference to the top ranked stocks. The performance of the model proved to be satisfactory for short-term investment period when compared with the recent performance of the stocks. The model is also proved to be more efficient and robust when compared with other existing expert system models for stock selection and portfolio recommendation.

Keywords: Ranking of stocks; Stock Portfolio Selection; Fuzzy rule-based system; Dempster-Shafer Theory; Fuzzy Delphi method.

1. Introduction

Stock portfolio selection is basically an optimal allocation of investor' assets among certain number of stocks to provide maximum return with minimum risk. There are different types of factors which directly or indirectly influence the performance of stocks. This makes the task of stock selection and portfolio construction challenging and uncertain for the stock market researchers since the beginning. According to the work of Markowitz [1] portfolio selection may include two stages: firstly, performances of different securities is observed with beliefs about their future performances and secondly, a proper choice of portfolio is made with relevant beliefs about future performances. The main aim in Modern Portfolio Theory (MPT) of investment is to maximize the expected return of the portfolio for a given amount of portfolio risk or equivalently minimize risk for a given level of expected return, by carefully choosing the proportions of various assets. In the ground-breaking work, Markowitz [1] has quantified return as the mean and variance as the risk of the portfolio of securities. The relationship between risk and mean-variance are later examined in various researches [2,3,4,5]. The twin objectives of investors', profit maximization and risk minimization, are thus quantified.

Portfolio selection problem is proved to be a multi-dimensional problem and to solve this inherent multi-criteria nature of this problem Multi-criteria Decision Making (MCDM) approach is adopted in various researches like [6,7,8,9]. In spite of all these efforts researchers were facing difficulties in bringing efficiency in portfolios specially in dynamic uncertain environment. As a result the use of different AI techniques in stock selection and portfolio construction became inevitable due to their well known capability of handling uncertain and vague data. In this article a DS theory based fuzzy expert system model is proposed.

The rest of the paper is organized as follows: section 2 gives a brief literature review where some significant researches, applying different AI and soft computing techniques in stock selection and portfolio recommendation, are introduced. In section 3 the proposed approach and its application is demonstrated. Selection of four critical factors and their importance in stock performance are discussed in section 4. In section 5 the concepts DS evidence theory and fuzzy expert system are briefly discussed. In section 6 the process of collecting historical data from BSE and its fuzzification process is explained. Fuzzy rule construction using DS theory is elaborated in section 7. In section 8 ranking of of stocks under BSE are demonstrated. Portfolio optimization process is elaborated in section 9. Results of the proposed model are analyzed and compared in section 10. Finally section 11 gives the conclusion.

2. Literature Review

In literature we can find wide use of different AI and Soft Computing techniques to analyse and predict the performance of stocks in various stock exchanges around the world. In some researches [10,11,12,13,14] efficient learning capability of Artificial Neural Network (ANN) is adopted to select stocks and construct portfolios whereas in some other researches [15,16,17,18,19] optimization capability of Genetic Algorithm is used for portfolio optimization. But the application of Fuzzy Logic and Fuzzy Set theory has become the most popular among other soft computing techniques, due to its uncertainty handling capability and the efficiency to incorporate vagueness in investors' preferences in portfolio construction.

In a research [20] proposed a new method regarding fuzzy MCDM by making some modification of Chen's Method [21] and applied it to the stocks selection on Istanbul Stock Exchange (ISE). In another article [22] applied fuzzy set theory to the problem of portfolio selection using Sharpes single-index model as a basis. In another article a portfolio rank-index is proposed by [23] that allows us to rank a finite set of portfolios based on the well-known value-ambiguity indices of trapezoidal fuzzy numbers. In another research [24] described a new method for design of fuzzy expert system for portfolio recommendation at Taiwan stock exchange (TSE).

3. Proposed Approach and its Application

Portfolio selection process can be divided into two stages [25]. In the first stage, some stocks are selected to construct the portfolio and in the second stage, the percentage of the total value for each stock is identified. Use of fuzzy expert systems in the selection of stocks and equity have recently become very effective and popular. In a work Fasanghari et al. [24] developed an fuzzy expert system to select superior stocks in Tehran Stock Exchange (TSE) by identifying 7 factors influencing the stock market. A rule base of total 932 rules was constructed for this purpose. Though no proper portfolio optimization model was used in this work, the outcome of the model proved to be satisfactory. But the major concern of this model can be the development time and cost due to repetitive expert interactions for the development of the model. Authors have used only fuzzy set theory to deal with the inherent uncertainty present in the rule base though fuzzy set theory is more effective in dealing with vagueness in the model than addressing inherent uncertainty of the model [26]. In another significant work Xidonas et al.[25] developed an expert system for the selection of equity securities but the concerns remained more or less same as the previous one.

In this research selection of stocks are made with the help of an expert system where the fuzzy rule base is automatically generated by applying DS evidence theory.

- This has minimized the overall development time and cost as the required number of expert interactions is reduced considerably.
- DS evidence theory is well known for its capability of dealing with incomplete and uncertain information. So another level of uncertainty handling mechanism beside fuzzy set theory is incorporated in the model.

The proposed model is implemented in two phases.

(1) Initially 10 investment ratios are selected as critical factors. Then the number of factors are reduced to 6 by applying fuzzy Delphi method. After calculating correlation coefficients among these factors, 4 factors with minimum correlations are finally selected and their historical data for the period of FY 2003-04 to FY

2012-13 are collected. These four factors have functioned as the input of the proposed DS fuzzy expert system and their historical data have been used to generate the consequents of the fuzzy rules with the help of DS evidence theory. The model is tested with the historical data of FY 2012-13, when values of these four factors for each stocks are provided as input to the fuzzy inference system and then they are ranked based on their defuzzified output.

(2) A portfolio optimization model is constructed and then it is solved with ACO by considering top 10 stocks, as per the previous ranking, in the portfolio, to allocate investment ratios by giving preferences to the top ranked stocks.

The model is tested with the data of FY 2012-13 to predict the performance of different stocks in FY 2013-14, FY 2014-15 and FY 2015-16. Performance of the model is proved to be satisfactory when compared with the actual performance of stocks. Figure 1 depicts the brief structure of the article.

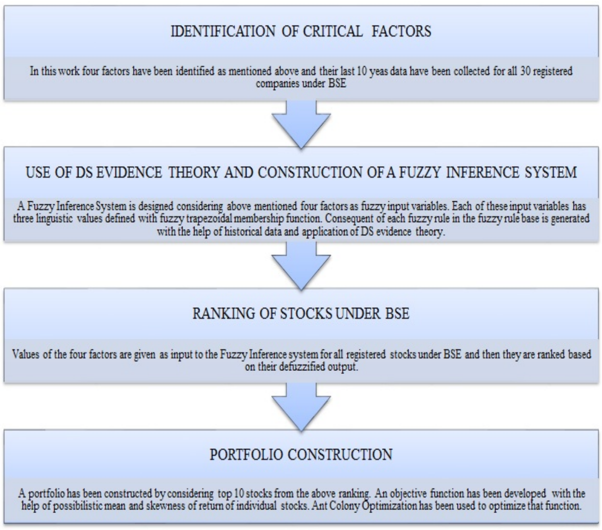

Fig. 1. Design diagram of the proposed inflation forecasting model

4. Identification of Critical Factors

According to value investors, there is no *right way* to analyze stocks due to the presence of multi-dimensional uncertainties. Stock market experts in BSE use many factors for the evaluation and selection of stocks. With the help of various experts' opinions and thorough literature survey, initially following 10 metrics (ratios) are identified.

Price to earnings (P/E)

The price to earnings ratio (P/E) is a valuation ratio of a stock that is determined by market value per share divided by earnings per share (EPS) of the stock.

$$\frac{P}{E} = \frac{\text{Market value per share}}{EPS}$$

In general, a high price to earnings ratio for any company indicates that investors are expecting higher earnings growth in future from it compared to companies with a lower price to earnings ratio.

Payout ratio

The payout ratio can be expressed as dividends paid out as a proportion of cash flow. It can also be expressed as the proportion of earnings paid out as dividends to shareholders, typically expressed as percentage.

$$Payoutratio = \frac{\text{Total dividends paid}}{\text{Net income}} \times 100$$

Earning per share(EPS)

It is the portion of a company's profit allocated to each outstanding share of common stock. Earning per share serves as indicator of a company's profitability and it is calculated as:

$$EPS = \frac{\textit{Net income} - \textit{Dividends on preferred stock}}{\textit{Average outstanding of shares}}$$

Price-to-book value (P/B)

Price-to-book vale (P/B) is the ratio of market value of a company's shares over its book value of equity.

$$\frac{P}{B} = \frac{\text{Stock price}}{\text{Total asset} - \text{Liabilities}}$$

Price to sales (P/S)

The price to sales ratio is another valuation ratio that compares a company's stock price to its revenues. It can be calculated by dividing the company's market capitalization by its total sale. This ratio is also known as revenue multiple or sales multiple.

$$\frac{P}{S} = \frac{\text{Market Capitalization}}{\text{Total Sales}}$$

Long term debt/equity (LTDER):

The long term debt to equity ratio is the ratio of total liabilities of a stock to its shareholders' equity. It is a leverage ratio and it measures the degree to which the assets of the stock are financed by the debts and the shareholders' equity of a stock. It is calculated by the total liabilities divided by shareholders' equity.

$$LTDER = \frac{\text{Total Liabilities}}{\text{Shareholders' Equity}}$$

Price-to-cash-flow ratio (P/CF)

The price-to-cash-flow ratio is a stock valuation indicator that measures the value of a stock's price to its cash flow per share.

$$\frac{P}{CF} = \frac{\text{Share price}}{\text{Cash flow per share}}$$

Current ratio (CR)

The current ratio measures a company's ability to pay short-term and long-term obligations.

$$CR = \frac{\text{Current assets}}{\text{Current liabilities}}$$

Profit margin (PM)

Profit margin is a profitability ratio calculated as net income divided by revenue, or net profits divided by sales.

$$PM = \frac{\text{Net income}}{\text{Net sales}}$$

Accounts receivable turnover (ART)

Receivables turnover ratio can be calculated by dividing the net value of credit sales during a given period by the average accounts receivable over the same period.

$$ART = \frac{\text{Net credit sales}}{\text{Average accounts receivable}}$$

To reduce the complexity in the proposed expert system number of factors needed to be reduced. A questionnaire survey is conducted to select most important factors from tacit knowledge of experts. The questions were about the importance of these 10 factors in stock selection and for that a 1–10 point scale is used. Higher importance is indicated by higher point. Though the questionnaire were distributed to 65 domain experts, 40 of them successfully completed the survey. To select the critical factors from this survey Fuzzy Delphi Method [27] is applied. Fuzzy Delphi method and its application in factor identification is discussed below.

4.1. *Fuzzy Delphi Method*

Traditional Delphi method was integrated with fuzzy set theory to improve the ambiguity and vagueness of the Delphi method [28] where membership degree is used to establish the membership function of each participant. Later to predict the prevalence of computers in the future, max–min and fuzzy integration algorithms were developed by introducing fuzzy theory into Delphi method [29]. Triangular fuzzy number is applied to Delphi method to incorporate expert opinion in another research [27]. The two terminal points of triangular fuzzy number represents the maximum and minimum values of experts' opinions and to derive the statistically unbiased effect and avoid the impact of extreme values, the geometric mean is taken as the membership degree of the triangular fuzzy numbers. In [30], this method is successfully implemented to construct key performance appraisal indicators for mobility of service industries. It is noticed from this research that besides its simplicity this model can encompass all the expert opinions in one investigation. Fuzzy Delphi method is also successfully applied in the determination of appraisal criteria for employees' performance evaluation based on MCDM technique [31]. The main advantage of this method in collecting group decision lies in that every expert opinion will be considered and integrated to achieve the consensus of group decisions [30]. Uncertain and subjective messages in human thinking can also be induced in this model. It also reduces the investigation time and cost.

For the selection of critical factors for stock evaluation fuzzy Delphi method proposed by [27] is applied in this research to denote expert consensus with geometric mean. The process is explained as follows:

4.1.1. *Representing Expert opinions in Triangular Fuzzy Number*

All expert opinions collected from questionnaire are organized into estimates and then the triangular fuzzy number $\widetilde{T}_F$ is created as follows:

$$\left.\begin{aligned} \widetilde{T}_F &= (L_F, M_F, U_F) \\ L_F &= min(X_{F_i}) \\ M_F &= \sqrt[n]{\prod_{i=1}^{n} X_{F_i}} \\ U_F &= max(X_{F_i}) \end{aligned}\right\} \tag{1}$$

where i denotes the ith expert, $i = 1, 2, 3, ..., n$ and X_{F_i} indicates evaluation value of the ith expert for factor F; The bottom of all experts' evaluation scores for factor F is represented by L_F; U_F indicates the ceiling of all the experts' evaluation scores for the factor F and M_F represents the geometric mean of all the experts' evaluation scores for the factor F.

4.1.2. *Selection of Factors*

The geometric mean M_F of each factor's triangular fuzzy number is used to denote the expert group consensus on the importance value of the 10 previously identified factors. This is explicitly done to avoid the impact of extreme values. Table 1 describes the geometric mean calculated for these 10 factors. The selection criteria for threshold value σ is decided as below:

$$\text{If } M_F \geq \sigma = 7, \text{ the factor is accepted.}$$
$$\text{If } M_F < \sigma = 7, \text{ the factor is rejected.}$$

Table 1. Factors with geometric mean

Sl No.	Factor	Geometric Mean (M_F)
1	Earn per share	7.17
2	Payout ratio	7
3	Price to earning ratio	8.43
4	Price to sales ratio	8.35
5	Current ratio	6.02
6	Price to book value ratio	8.26
7	Price to cash flow ratio	6.59
8	Profit margin	6.52
9	Long term debt to equity ratio	8.20
10	Accounts receivable turnover	5.81

So finally 6 factors, EPS, PR, P/E, P/S, P/B and LTDER satisfied the threshold criteria and selected for further consideration.

4.2. *Selection of Input Factors for the Proposed Model*

In this research fuzzy expert system is used for the selection and ranking of stocks. Consequents of the fuzzy rules in the knowledge base are automatically generated by applying DS theory where historical data of the few input factors are served as evidences of their performance. But one of the major constraints in DS evidence theory is that all evidences used are necessarily required to be conditionally independent. In stock market we could not find any significance evidence which can conclude whether the above 6 selected factors are conditionally dependent or not. So we have checked the dependencies of these factors by using their historical data of last three years (2012-13, 2013-14 and 2014-15). Semivariance to return ratio (S/R) is treated as one of the most effective and popular performance indicator of stocks in various stock exchanges. Value of any factor in any financial year is divided by the S/R value of that particular stock for the same year to determine the impact of a factor in the overall performance of the stock. This result is termed as 'impact score'. For example, the value of P/E for Axis Bank Ltd. was 17.11 and

S/R was 7.52 in FY 2014-15. So the impact score of P/E for Axis Bank Ltd. in FY 2014-15 is calculated as 2.27. Similarly for all 30 registered stocks under BSE, impact scores of all 6 previously selected factors are calculated for FY 2012-13, FY 2013-14 and FY 2014-15. Dependencies of these factors are determined through the following steps.

4.2.1. *Generation of Correlation Coefficient Matrix*

Degree of linear independence between two random variables can be measured by their correlation coefficients. If two variables U and V has N scalar observations each, then the Pearson correlation coefficient can be defined as

$$\rho(U,V) = \frac{1}{N-1}\sum_{i=1}^{N}\left(\frac{U_i-\mu_U}{\rho_U}\right)\left(\frac{V_i-\mu_V}{\rho_V}\right) \tag{2}$$

where μ_U and ρ_U are the mean and standard deviation of U, respectively and μ_V and ρ_V are the mean and standard deviation of the variable V. We can also define the correlation coefficient in terms of the covariance of U and V as follows

$$\rho(U,V) = \frac{cov(U,V)}{\rho_U\rho_V} \tag{3}$$

The correlation coefficient matrix of two random variables is generated as

$$M = \begin{pmatrix} \rho(U,U) & \rho(U,V) \\ \rho(V,U) & \rho(V,V) \end{pmatrix} \tag{4}$$

Considering 6 factors as random variables, the 30 registered stocks as 30 observations and corresponding impact scores as their values, correlation coefficient matrices are calculated using equation 4 for the three financial years. Table 2 shows these correlation coefficient values for FY 2014-15.

Table 2. Factors with geometric mean

Factors	P/E	P/B	P/S	LTDER	EPS	PR
P/E	1	0.57	0.051	0.36	0.19	0.25
P/B	0.57	1	0.17	-0.08	0.39	0.50
P/S	0.051	0.17	1	0.24	0.02	0.22
LTDER	0.36	-0.08	0.24	1	-0.10	-0.07
EPS	0.19	0.39	0.027	-0.10	1	0.85
PAYOUT	0.25	0.50	0.22	-0.07	0.85	1

Table 2 gives a hint about the dependencies of factors among each other. Higher dependencies are indicated by higher values and lower values indicate the lower dependencies. The level of dependencies of factors are shown in Figure 2, 3 and 4 for the three financial years.

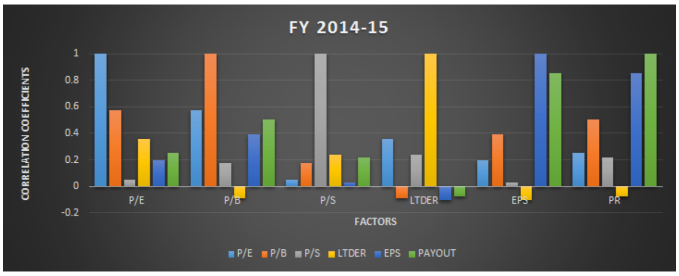

Fig. 2. Dependencies of factors in FY 2014-15

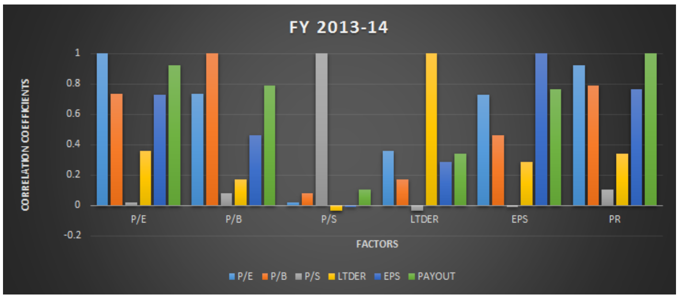

Fig. 3. Dependencies of factors in FY 2013-14

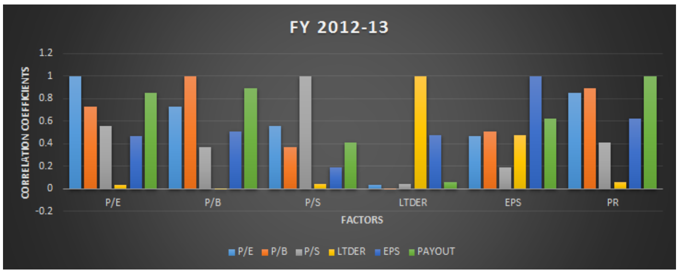

Fig. 4. Dependencies of factors in FY 2012-13

In all the three years it is noticed that correlation coefficient values are high among P/E and P/B. This gives an indication that P/E and P/B are highly dependent with each other. EPS and PR are also found to be highly dependent. So if P/E and P/B are used as evidences in DS evidence theory at the same time, it may lead to a result of super estimate. The same is true for EPS and PR. But the dependencies of LTDER, P/S, P/E (or P/E) and EPS (or PR) among each other are relatively low. So all of these factors can be used as evidences for the DS synthesis of the proposed fuzzy rule based system. In this research finally the historical values of P/E, P/S, EPS and LTDER are treated as input to the proposed DS fuzzy expert system and are hence used for fuzzy rule construction and ranking of the stocks.

5. Brief introduction to DS evidence theory and fuzzy expert system

DS evidence theory and fuzzy rule based expert system are hybridized in this work to evaluate and rank the stocks under BSE. Brief introduction of DS evidence theory and fuzzy rule based expert system are given here before their application in the proposed model is elaborated.

5.1. *DS evidence theory*

A. P. Dempster in 1967 [32] proposed a multi-valued mapping from one space to another which was used for statistical inference when we needed to identify a single hypothesis from multiple sample information. The DS-theory was first proposed by Dempster in 1968 [33] and later in 1976 it was extended by Shafer [34] to deal with incomplete and uncertain information.

DS-Theory mainly deals with four components: Frame of discernment, Basic Probability Assignment (BPA), plausibility function Pl and belief function Bel.

In evidence theory first a set of hypothesises Θ called frame of discernment, is assumed and defined as follows:

$$\Theta = \{H_1, H_2, H_3, ..., H_N\}. \tag{5}$$

The set is composed of N mutually exhaustive and exclusive hypothesises. From the Θ, let us denote $P(\Theta)$, be the power set composed with 2^N propositions $X$of Θ:

$$P(\Theta) = \{\varnothing, \{H_1\}, \{H_2\}, ..., \{H_N\}, \{H_1 \cup H_2\}, \{H_1 \cup H_3\}, ..., \Theta\} \tag{6}$$

where $\varnothing$ denotes the empty set. The most important task in applying evidence theory is Basic Probability Assignment (BPA). A BPA is a function from $P(\Theta)$ to $[0, 1]$ defined by:

$$\begin{aligned} m: \quad & P(\Theta) \to [0,1] \\ & X \mapsto m(A) \end{aligned} \tag{7}$$

and which satisfies the following criteria:

$$\begin{aligned} \sum_{X \in P(\Theta)} m(X) &= 1; \\ m(\varnothing) &= 1. \end{aligned} \tag{8}$$

Dempster's rule of combination is denoted by $m = m_1 \oplus m_2$, which can combine two BPAs m_1 and m_2 to yield a new BPA:

$$m(A) = \frac{\sum_{Y \cap Z = A} m_1(Y) m_2(Z)}{1 - k} \tag{9}$$

with

$$k = \sum_{Y \cap Z = \varnothing} m_1(Y) m_2(Z) \tag{10}$$

where k is a normalization constant, known as conflict because it measures the degree of conflict between m_1 and m_2. $k = 0$ corresponds to the absence of conflict between m_1 and m_2, whereas $k = 1$ implies total contradiction between them. The resulting belief function from the combination of I information sources can be defined as:

$$m = m_1 \oplus m_2 \oplus m_3 ... \oplus m_I \tag{11}$$

With the help of Equation (11) multi source information can be fused into a single framework in belief theory very easily and it proved to be very effective when it is applied in fuzzy rule generation discussed later in Section 3.3.3.

5.2. *Fuzzy Expert System*

For many real world problems, like strategic planning, medical diagnosis and other decision making problems solutions are something more than simple reasoning and require some expertise knowledge to solve. So the use of Expert Systems (ES) became popular because it is capable of representing and reasoning knowledge rich domain with a view to solve problems and giving advice [35]. The main advantage of ES over any other conventional software applications is its effective reasoning capability as they process knowledge instead of data or information [36]. On the other hand the application of fuzzy set theory in uncertainty handling has become one of the strongest tool to design any framework. As fuzzy set theory is now widely applied in information gathering, modeling, analysis, optimization, control, decision-making and supervision [37], fuzzy expert system is used in the construction of the proposed model.

Fuzzy expert system is an expert system that uses fuzzy logic instead of Boolean logics in its knowledge base and derives conclusion from user inputs and fuzzy inference process [38]. Figure 5 depicts the basic architecture of a fuzzy expert system.

Various components of a fuzzy expert system are briefly discussed below:

- **Fuzzy Knowledge Base**: Fuzzy knowledge base is a collection of fuzzy rules and it is built up by the knowledge engineer after eliciting knowledge from domain experts. The basic structure of a fuzzy rule is as follows:

  IF I_1 is i_1 AND I_2 is i_2 AND ... AND I_n is i_n THEN O is o.

  where I_1, I_2 ... , I_n are fuzzy linguistic input variables and i_1, i_2, ... , i_n are possible linguistic values of I_1, I_2 ... , I_n respectively. Similarly, O is fuzzy linguistic output variable and o is its linguistic value. The terms I_1 is i_1 and I_2 is i_2 ... are fuzzy propositions suggesting partial set membership or partial truth. In this way modifying the fuzzy sets parameters of the output fuzzy sets, the partial truth of the rule premise can be evaluated [39].

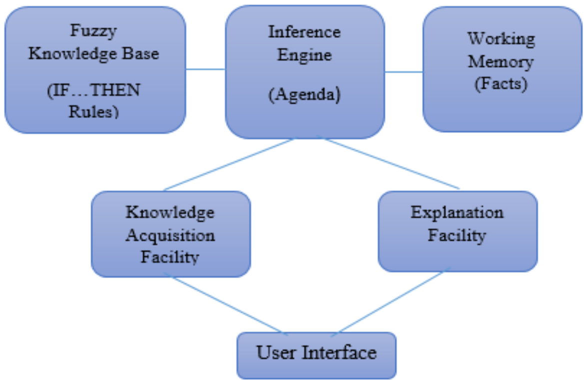

Fig. 5. Architecture of Fuzzy Rule-based Expert System

- **Fuzzy Inference Engine**: To analyse and manipulate the rules in the knowledge base an inference mechanism is adopted to conclude a logical result. Various inference mechanisms can be adopted for fuzzy inference engine depending on the aggregation, implications and operators used for s-norms and t-norms [40]. Defuzzification of the results is also achieved in this module. This modules functions as per the agenda. An agenda is a prioritized list of rules identified and enlisted by inference engine. Patterns of these rules are satisfied by facts or objects in the working memory.
- **Working Memory**: This module is contained of a global database of facts those are going to be used by the fuzzy rule base.
- **Explanation Facility**: This particular module explains every steps in reasoning process of the ES.
- **Knowledge Acquisition Facility**: This module provides an automated facility for the user to enter knowledge in the system.
- **User Interface**: All kinds of interactions of the user with the system is facilitated by this module. This is used to accept inputs from the user and display the outcomes of the inference process.

6. Collection of historical data and its fuzzification

Selection of input factors are discussed in details in section 4. Next stage of the development process is fuzzification of the historical data collected for these factors.

6.1. *Collection of historical data*

Total 30 stocks are registered under BSE. Initially, historical data of last 12 years (2003-04 to 2014-15) of four factors (P/E, EPS, P/S and LTDER) are collected from different web sources like www.capitaline.com, www.bseindia.com, www.nseindia.com etc. First 9 years' data are used for the implementation of the model and last 3 years' data are used for testing and comparison. For the simplicity of the model the data are normalized within the range of [0, 10] by scaling up/down the maximum historical value of any factor in last 9 years to 10 and then scaling other data accordingly relative to the maximum value. Table 3 shows an example of actual and normalized data for Axis Bank Ltd.

Table 3. Sample normalized data for Axis Bank Ltd.

Financial Year	Actual P/E ratio of Axis Bank Ltd.	Transferred Data
2011-12	11.46	4.24
2010-11	17.5	6.48
2009-10	19.47	7.2
2008-09	8.49	3.14
2007-08	**27.02**	**10**
2006-07	21.67	8.02
2005-06	21.12	7.82
2004-05	20.5	7.59
2003-04	12.54	4.64

6.2. *Fuzzification*

Fuzzification is a process of taking a crisp value as input and transforming it into the degree required by the terms. Uncertainty basically comes from imprecision, vagueness or ambiguity and one of the best ways to represent these in any variable can be in the form of fuzzy variable. Fuzzy membership functions are used to represent the fuzzy variables. All four factors have been illustrated in Table 4 in terms of fuzzy linguistic variables and each of them has three linguistic values, *Low, Standard and High.* Trapezoidal membership function has been used to represent the inputs within the range of [0, 10].

7. Fuzzy Rule Construction using DS-Theory

The most significant novelty of this research work is the application of DS evidence theory in fuzzy rule construction. The knowledge base of any fuzzy inference system is developed using collection of fuzzy rules which determines how the output will be generated based on the given input. The proposed system has four input parameters namely, P/E, EPS, P/S and LTDER, and based on these input parameter the DS-Fuzzy system is expected to determine whether the selection of any stock will be *Highly Favourable, Moderately Favourable or Not Favourable*, as output. The frame of discernment is considered as $\Theta = \{High\ Performance,\ Average\ Performance,\ and\ Poor\ Performance\}$.

Risk/Return ratio is one of the most commonly used measure to evaluate the performance of stocks. Risk is generally measured by semi-variance(S) of the stocks'

Table 4. Membership functions for the linguistic values of the Input Variables

Factors in terms of linguistic variables	Linguistic Values	Fuzzy Trapezoidal Membership
P/E Ratio (Range 0 to 10)	Low	(0 0 1.8 2.8)
	Standard	(1.7 3.5 4.6 5.8)
	High	(5.3 7.5 10 10)
EPS (Range 0 to 10)	Low	(0 0 2.2 3.5)
	Standard	(2.5 4.6 5.6 7.9)
	High	(6.4 9.6 10 10)
P/S Ratio (Range 0 to 10)	Low	(0 0 2.4 3.6)
	Standard	(1.8 4.4 5.7 8.2)
	High	(6.4 8.7 10 10)
LTDER (Range 0 to 10)	Low	(0 0 2.3 3.6)
	Standard	(2.8 4.6 5.7 7.6)
	High	(6.124 8.09 10 10)

previous returns and Return(R) is actually the mathematical mean of previous returns. Obviously lower value of this ratio indicate better performance of stocks. Based on linguistic values of each input factors and comparing those with S/R values as a measure of the performance of various stocks BPAs for different hypothesises are achieved. The historical data of FY 2012-13 are used to identify the stocks. Consulting various domain experts and analysing historical data initially standard values for each factors for all the stocks are decided. By analysing last 9 years' historical data it is found that when P/E ratio was around its standard value then 75% of stocks under BSE performed better and S/R value was also found to be satisfactory. So a belief of 0.75 is assigned towards the hypothesis $\{High\ Performance\}$ when the P/E ratio is near to its standard value. Again, when P/E value found to be much lower than its standard value, 65% of total stocks under BSE performed average and S/R value also found to be average. So 0.6 degree of belief is assigned towards the hypothesis $\{Average\ Performance\}$ when P/E ratio is lower than its standard value. Similarly, when the value of P/E ratio was much higher than the standard value then it was found that 70% of stocks under BSE performed poor and S/R value was also very high. So 0.7 degree of belief is assigned towards the hypothesis $\{Poor\ Performance\}$. In the same way initial believes or Basic Probabilities are assigned for all other factors as mentioned in Table 5.

Now to elaborate further let us consider the following four sample rules with their initial belief:

(1) IF P/E is **Low** THEN Performance will be **Average** ($m_1\,(A\_P) = 0.6$).
(2) IF EPS is **Standard** THEN Performance will be **Average** ($m_2\,(A\_P) = 0.6$).
(3) IF P/S is **High** THEN Performance will be **High** ($m_4\,(H\_P) = 0.75$).
(4) IF LTDER is **High** THEN Performance will be **Poor** ($m_6\,(P\_P) = 0.65$).

The above four rules represent the belief towards the performance of stocks with respect to the values of four input factors as the presence of evidences. Now consider the IF part of Rule 1 and Rule 2 as the first two evidences and m_1 and m_2 as mass

Table 5. Basic Probability Assignment

Factors	Linguistic Values	High Performance (H_P)	Average Performance (A_P)	Poor Performance (P_P)
	Low		0.6	
P/E	Standard	0.75		
	High			0.7
	Low			0.8
EPS	Standard		0.6	
	High	0.65		
	Low			0.7
P/S	Standard		0.65	
	High	0.75		
	Low	0.6		
LTDER	Standard		0.75	
	High			0.65

functions for them respectively.

Now from Rule 1, $m_1(A\_P) = 0.6$ and $m_1(\Theta) = (1 - 0.6) = 0.4$, where, $m_1(\Theta)$ represents the degree of belief in the rest of the hypotheses present in the hypothesis set. Similarly, from Rule 2, $m_2(A\_P) = 0.6$ and $m2(\Theta) = (1-0.6) = 0.4$. Now these two mass functions are combined using Dempster's rule of combination (Equation 9) to obtain a new function m_3 as mentioned in Table 6.

Table 6. Combination of mass considering first two evidences

Combining m_1 and m_2	$m_2(A\_P) = 0.6$	$m_2(\Theta) = 0.4$
$m_1(A\_P) = 0.6$	$A\_P = 0.36$	$A\_P = 0.24$
$m_1(\Theta) = 0.4$	$A\_P = 0.24$	$\Theta = 0.16$

The new mass value m_3 for the same hypothesis set is calculated as:

$$\left.\begin{aligned} m_3(A\_P) &= 0.36 + 0.24 + 0.24 = 0.84 \\ m_3(\Theta) &= 0.16 \end{aligned}\right\} \quad (12)$$

Now consider the IF part of the Rule 3 as the new evidence and m_4 be the new mass function. From Rule 3, $m_4(H\_P) = 0.75$ and $m_4(\Theta) = (1 - 0.75) = 0.25$. Now again, m_3 and m_4 are combined to generate new mass m_5 as mentioned in Table 7.

New mass values m_5 for the hypothesis set are calculated as below:

$$\left.\begin{aligned} m_5(A\_P) &= \tfrac{0.21}{1-0.63} = 0.567 \\ m_5(H\_P) &= \tfrac{0.12}{1-0.63} = 0.324 \\ m_5(\Theta) &= \tfrac{0.04}{1-0.63} = 0.108 \end{aligned}\right\} \quad (13)$$

Table 7. Combination of mass considering first three evidences

Combining m_3 and m_4	$m_4(H\_P) = 0.75$	$m_4(\Theta) = 0.25$
$m_3(A\_P) = 0.84$	$\Phi = 0.63$	$A\_P = 0.21$
$m_3(\Theta) = 0.16$	$H\_P = 0.12$	$\Theta = 0.04$

Now if we consider the IF part of the Rule 4 as the last evidence and m_6 as its mass function then we get $m_6(H\_P) = 0.6$ and $m_6(\Theta) = (1 - 0.6) = 0.4$. Now finally, m_5 and m_6 are combined as mentioned in Table 8 to generate the final mass values m_f for the hypothesis set (Equation 11).

Table 8. Combination of mass considering all four evidences

Combining m_5 and m_6	$m_6(P\_P) = 0.65$	$m_6(\Theta) = 0.35$
$m_5(A\_P) = 0.567$	$\Phi = 0.368$	$A\_P = 0.198$
$m_5(H\_P) = 0.324$	$\Phi = 0.21$	$H\_P = 0.113$
$m_5(\Theta) = 0.108$	$P\_P = 0.07$	$\Theta = 0.037$

$$\left.\begin{aligned} m_f(H\_P) &= \tfrac{0.113}{1-0.578} = 0.267 \\ m_f(A\_P) &= \tfrac{0.198}{1-0.578} = 0.469 \\ m_f(P\_P) &= \tfrac{0.07}{1-0.578} = 0.165 \\ m_f(\Theta) &= \tfrac{0.37}{1-0.578} = 0.087 \end{aligned}\right\} \tag{14}$$

Following the same procedure final mass values for the rest 80 rules are calculated. For the hypothesis *High Performance(H_P)* maximum and minimum mass values (m_f) are found to be 0.9916 and 0 respectively. As mass values for any hypothesis can lie between 0 and 1 we have divided the favourability of any stock, based on the $m_f(H\_P)$ of the rules, into three categories: *Highly Favourable* ($0.76 \leq m_f(H\_P) \leq 1$), *Moderately Favourable* ($0.46 \leq m_f(H\_P) \leq 0.75$) and *Not Favourable* ($m_f(H\_P) \leq 0.45$).

So the combined version of the four sample rules mentioned earlier is formed as: "IF P/E is **Low** AND EPS is **Standard** AND P/S is **High** AND LTDER is **High** THEN the stock is **Not Favourable**".

In this way total 81 fuzzy rules are formulated using DS evidence theory to develop the knowledge base of the proposed DS-fuzzy inference system and the above mentioned three selection categories are converted into fuzzy linguistic variables with trapezoidal membership values as shown in Table 9 for output of the proposed inference system.

Table 9. Degree of membership of linguistic values for the output variable

Output Variable	Linguistic Values	Fuzzy Trapezoidal Membership
Selection (Range: 0 to 1)	Not Favourable	(0 0 0.172 0.448)
	Moderately Favourable	(0.34 0.46 0.57 0.75)
	Highly Favourable	(0.64 0.88 1 1)

Figure 6 shows the snapshot of the fuzzy rule base and Figure 7 shows the sensitivity analysis of the same when designed in MATLAB.

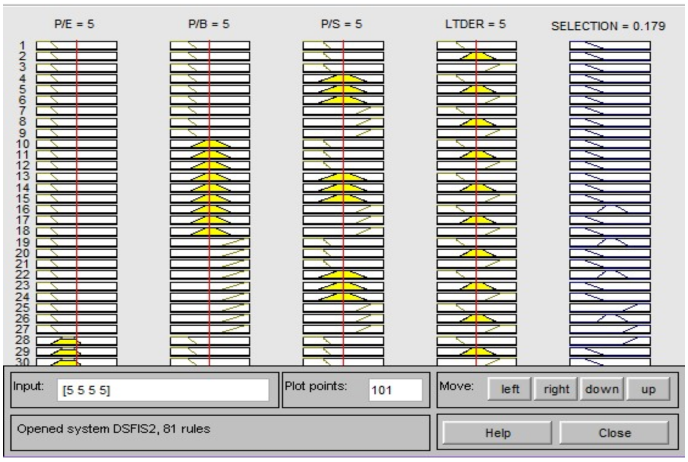

Fig. 6. Fuzzy rule base of the DS-fuzzy system for Stock Selection

8. Defuzzification and ranking of stocks

The process of obtaining a single value that represent the best outcome of the fuzzy set evaluation is known as defuzzification [41]. Though there are many other popular defuzzification methods namely, Bi-sector, Mean of Maximum (MOM), Smallest of Maximum (SOM), Largest of Maximum (LOM), in this proposed model ‘Centroid’ is used for the defuzzification purpose due to its wide range of acceptability. This method was introduced by Sugeno in 1985 and it can be expressed as:

$$z^* = \frac{\int \mu_{\underset{\sim}{C}}(z) \cdot z dz}{\int \mu_{\underset{\sim}{C}}(z) dz} \tag{15}$$

Where z is the output variable, z^* is the defuzzified output value and $\mu_{\underset{\sim}{C}}(z)$ is the aggregated membership function.

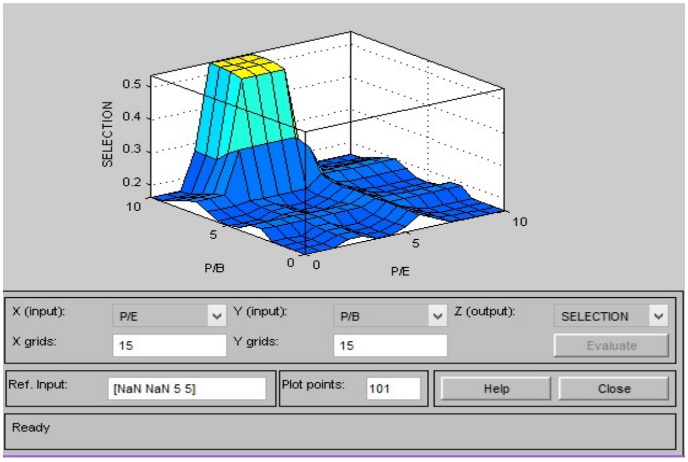

Fig. 7. The Sensitivity analysis of the DS-fuzzy system for Stock Selection

The proposed DS-fuzzy system is used to rank all 30 registered stocks under BSE based on the defuzzified outputs when the values corresponding to each of the four input factors for 2012-13 for each stocks are provided as input to the system. Top 10 stocks from this ranking are mentioned in Table 10.

Table 10. Top 10 Stocks based on the input data for FY 2012-13

Sl. No	Name of the Stocks	Defuzzified Values(FY 2012-13)
1	Hindustan Unilever Ltd.	0.8644
2	Sun Pharmaceutical Inds. Ltd.	0.8457
3	ITC Ltd.	0.5369
4	Coal India Ltd.	0.534
5	Tata Consultancy Services Ltd.	0.4936
6	Infosys Ltd.	0.3842
7	Dr. Reddy's Laboratories Ltd.	0.3553
8	Bajaj Auto Ltd.	0.3292
9	Hero Motocorp Ltd.	0.3002
10	Cipla Ltd.	0.285

As the consequent of every rule indicates the favourability of the stocks, higher defuzzified value indicate higher favorability. The highest defuzzified value of 0.8691 is found for *Hindustan Unilever Ltd.* among all 30 registered stocks and it has topped the ranking. These 10 stocks are used for portfolio construction as discussed in the next section.

9. Portfolio Construction

The first phase of stock portfolio selection, i.e. the selection of suitable stocks, is achieved in the previous section. In this section a portfolio construction model is discussed to determine optimum investment ratios for the selected securities such that the overall return is maximized under a tolerable risk.

9.1. *Objective Function*

The ratio of the difference of fuzzy portfolio return and the risk free return to weighted mean semi-variance of the assets is used as the objective function in this work. Certainly, higher value of this ratio will indicate the better investment; so the optimization target will be to maximize the ratio.

The notations used for the construction of this objective function are mentioned below:

- x_i: Fraction of the total investment allotted to the i^{th} asset;
- $\tilde{r_i}$: Fuzzy return of the i^{th} asset;
- r_f: Risk free return rate;
- μ_s: Weighted mean of asset semi-variances;
- r_p: Portfolio return;
- v_p: Variance of the portfolio;
- s_p: Skewness of the portfolio;

Thus the objective function is formed as:

$$Maximize \frac{E(\sum \tilde{r_i} x_i) - r_f}{\mu_s} \tag{16}$$

Where, after descending sort of portfolio, $\mu_s = \sum x_i s_i$, i.e. x_i is the i^{th} weight in the descending order and s_i is the semi-variance of the i^{th} ranked asset.

A fuzzy aggregation is used to calculate the fuzzy returns of the securities from the statistical database of past five years(2008-09 to 2012-13). If t_i means i^{th} position of the data, the fuzzy return can be calculated as:

$$\tilde{r_i} = \left(min(r_i), \frac{\sum t_i r_i}{\sum t_i}, max(r_i) \right) \tag{17}$$

The constraints included in this model are:

$$\left. \begin{array}{l} r_p > \alpha, v_p > \beta, s_p > \gamma \\ \sum\limits_{i=1}^{n} x_i = 1, x_i \leq M, x_i > 0, \forall i \end{array} \right\} \tag{18}$$

Values for α,β, γ, M and m are decided based on the investor's preferences.

Thus the final portfolio optimization model can be summarized as below:

$$\left.\begin{array}{l} Maximize \frac{E(\sum \tilde{r}_i x_i) - r_f}{\mu_s} \\ \text{Subject to,} \\ r_p > \alpha, v_p > \beta, s_p > \gamma \\ \sum_{i=1}^{n} x_i = 1, x_i \leq M, x_i > 0, \forall i \end{array}\right\} \quad (19)$$

9.2. *Optimization of the proposed model using ACO*

An algorithm using Ant Colony Optimization (ACO) is proposed and implemented in this section to solve the optimization model as mentioned in Equation (19). ACO is a very popular meta-heuristic optimization technique inspired by the foraging behaviour of biological ants [42,43]. Pseudo code of the proposed algorithm is given as below:

To execute the above algorithm in this work, initially 2000 random solutions are generated for the ant colony of 50 ants. Total 400 iterations are used for the optimization purpose while lifetime for each ant is considered as 20.

In Equation (19), $\tilde{r_i}$ is represented as a triangular fuzzy number. The Expected Return (E), Variance (Var), Skewness (Skew) and Semi-variances of these top 10 securities for past five years (2008-09 to 2012-13) as used in the above algorithm are calculated by the following theorem and are mentioned in Table 11.

Theorem 1. *Let $\tilde{A} = (a, b, c)$ be a triangular fuzzy number. The the weighted possibilistic mean, variance and skewness can be calculated as* [44]*:*

$$\left.\begin{array}{ll} E(\tilde{A}) & = \frac{1}{6}(a + 4b + c) \\ Var(\tilde{A}) & = \frac{1}{18}(a^2 + b^2 + c^2 - ab - bc - ca) \\ Skew(\tilde{A}) = & \frac{19(a^3+c^3)-8b^3-42b(a^2+c^2)+12b^2(a+c)-15(a^2c+ac^2)+60abc}{10\sqrt{2}(\sqrt{a^2+b^2+c^2-ab-bc-ca})^3} \end{array}\right\} \quad (20)$$

When the algorithm is executed in MATLAB with the above dataset and considering other parameters as $r_f = 0.01$, $\beta = 0.5$, $\alpha = 0.05$, $\gamma = 0.001$, $M = 0.8$ and $\mu_s = 0.0016$, the maximum return is found as 0.1317. The proposed ratio allocation for this return is given in Table 12.

The convergence of the objective values as per the proposed model is depicted in Figure 8 and the accumulation of ants to the optimum objective values in each iteration is depicted in Figure 9.

10. Result Analysis

In this section we have analyzed the performance of our proposed model in three different aspects.

Algorithm 1 ACO algorithm for portfolio optimization

```
procedure ACO-Portfolio
    Generate N random solution nodes based on Equation (19);
    Initialization of the ACO;
    for ITERATION=1 to I do
        for ANT=1 to Γ do
            Select the start node randomly;
            for LIFETIME=2 to L do
                Select the next node based on the heuristic information and
                pheromone concentration in the path. Move to the next node only
                if it is better than the current node;
                Update pheromone on the selected path;
            end for
            Store the details of the final node reached by each ants;
        end for
        Identify the solution node where maximum number of ants reached
        and consider that to be the optimum solution for the current solu-
        tion;
        Update the pheromone on the path of each ants who have reached this
        optimum solution;
        Evaporate the pheromone from all paths.
    end for
end procedure
```

Table 11. Expected Return, Variance, Skewness and Semivariance of stocks

Rank	Name of the Stocks	Return	Variance	Skewness	Semivariance
1	Hindustan Unilever Ltd.	0.1443	0.0001	0.2391	0.00004
2	Sun Pharmaceutical Inds. Ltd.	0.0529	0.0006	-0.3269	0.00254
3	I T C Ltd.	0.2801	0.0035	0.7523	0.00314
4	Coal India Ltd.	0.1275	0.0028	0.6046	0.0373
5	Tata Consultancy Services Ltd.	0.1228	0.0013	-0.5029	0.00557
6	Infosys Ltd.	0.0356	0.0003	0.8648	0.0002
7	Dr. Reddy'S Laboratories Ltd.	0.0484	0.0006	-0.0065	0.00098
8	Bajaj Auto Ltd.	0.0579	0.001	0.7262	0.0013
9	Hero Motocorp Ltd.	0.1379	0.0004	0.8057	0.00049
10	Cipla Ltd.	0.0437	0.0002	0.8258	0.00018

10.1. *Adaptability of the proposed model*

Initially a ranking based on the S/R values of the stocks for the year 2012-13 is prepared and then a portfolio with top 10 stock from this ranking is constructed

Table 12. Ratio allocation for the proposed portfolio

Hindustan Unilever Ltd.	Sun Pharma. Inds. Ltd.	ITC Ltd.	Coal India Ltd.	TCS Ltd.	Infosys Ltd.	Dr. Reddy's Lab. Ltd.	Bajaj Auto Ltd.	Hero Motocorp Ltd.	Cipla Ltd.
0.2403	0.2235	0.1970	0.0764	0.0574	0.0525	0.0452	0.0413	0.0407	0.0255

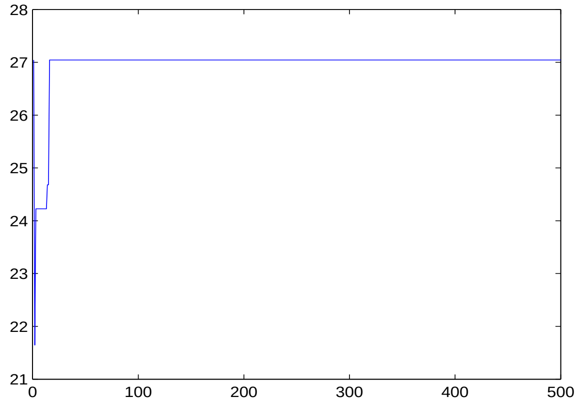

Fig. 8. Convergence of objective values based on proposed ranking

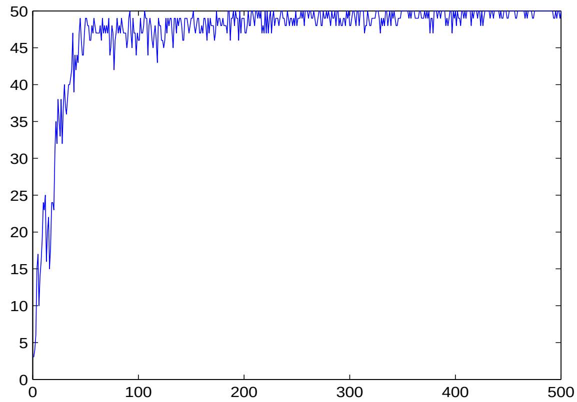

Fig. 9. Ant accumulation at optimum solutions

using the same objective value and ACO algorithm.

Figure 10 shows how the objective values are converged and Figure 11 displays the accumulation of ants at optimum nodes in each iterations.

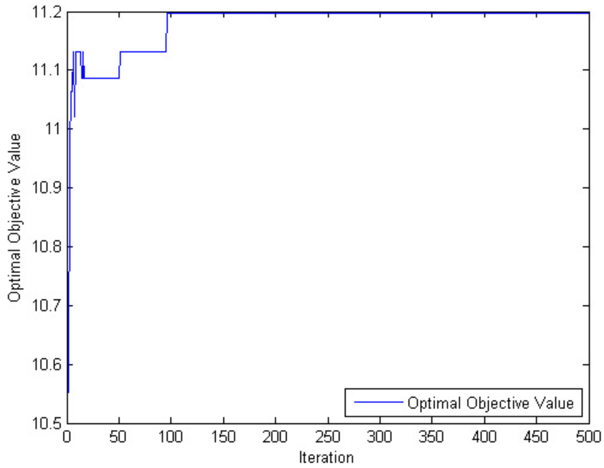

Fig. 10. Convergence of objective values

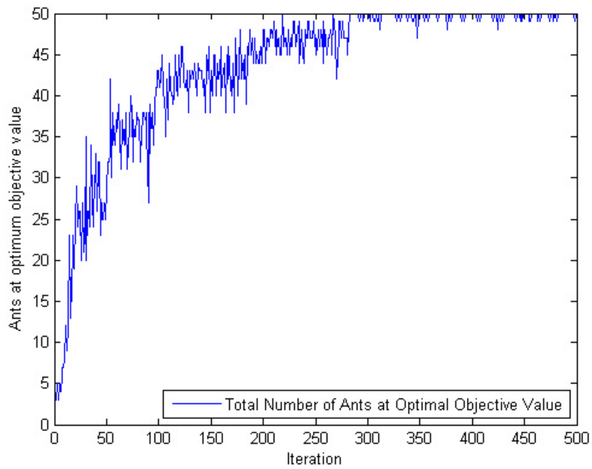

Fig. 11. Ant accumulations at optimum solutions

Return and risk of these two portfolios are compared in Table 13 and graphical representation of this comparison is shown in Figure

It is noticed that portfolio constructed based on the proposed ranking is capable of giving better return under comparatively lower risk. This outcome assures the adaptability of the proposed model by ensuring better portfolio return for short term investment period.

Table 13. Ratio allocation for the proposed portfolio

Type of the portfolio	Portfolio Return ($\sum \tilde{R}_i x_i$)	Portfolio Risk (μ_s)	Risk-Return Ratio
Based on proposed ranking	0.1316	0.0044	0.0334
Based on ranking using S/R values	0.0740	0.0057	0.0772

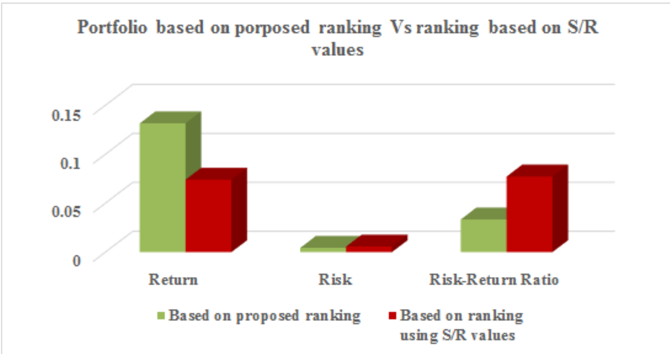

Fig. 12. Portfolio based on proposed ranking and based on ranking using S/R values

10.2. *Stability and effectiveness of the proposed model*

To check the reliability of the proposed ranking data are collected for next three financial years (FY 2013-14, FY 2014-15, FY 2015-16) and then stocks are ranked in risk-return framework. When these ranking are compared with the predicted top 15 stocks using the proposed model a match for 10, 11 and 9 companies are found in FY 2013-14, FY 2014-15 and FY 2015-16 respectively.

Presence of similar stocks in the above rankings promotes the effectiveness and stability of the proposed model for short and mid term investment periods.

10.3. *Comparison of other existing expert systems with the proposed model- An empirical study*

In dynamic uncertain environment selection of suitable stocks from any stock market for portfolio construction is always an challenging issue. In section 1 we have mentioned few researches which have addressed this challenging issue. In this article we propose a DS theory based fuzzy expert system to address the challenges of stock selection and portfolio construction. In literature we found four experts systems prior to these this work to address these challenging issues. These expert systems are implemented in different stock exchanges using different input factors and different fuzzy inference engines. So, we compared our proposed model with these four expert systems empirically. Table 15 shows details of these comparison.

Fuzzy rule base of previous expert systems were mainly designed based on ex-

Table 14. Top 15 stocks under BSE

Rank	Top 15 based the proposed model	Top 15 based on performance (2013-14)	S/R	Top 15 based on performance (2014-15)	S/R	Top 15 based on performance (2015-16)	S/R
1	Hindustan Unilever Ltd.	Hero Motocorp Ltd.	0.035	ITC Ltd.	0.019	TCS Ltd.	0.034
2	Sun Pharmaceutical Inds.	ITC Ltd.	0.043	HDFC Bank Ltd.	0.062	ITC Ltd.	0.042
3	ITC Ltd.	Hindustan Unilever Ltd.	0.081	Hero Moto Corp Ltd.	0.116	Hindustan Unilever Ltd.	0.056
4	Coal India Ltd.	Hindalco Industries Ltd.	0.087	Hindustan Unilever Ltd.	0.122	Tata Power Co. Ltd.	0.064
5	TCS Ltd.	HDFC Ltd.	0.123	Wipro Ltd.	0.142	State Bank Of India	0.114
6	Infosys Ltd.	TCS Ltd.	0.130	Maruti Suzuki India Ltd.	0.162	HDFC Bank Ltd.	0.117
7	Dr. Reddy's Laboratories Ltd.	HDFC Bank Ltd.	0.146	Tata Power Co. Ltd.	0.167	Wipro Ltd.	0.140
8	Bajaj Auto Ltd.	Tata Power Co. Ltd.	0.166	Tata Motors Ltd.	0.177	ICICI Bank Ltd.	0.191
9	Hero Moto Corp Ltd.	Coal India Ltd.	0.180	TCS Ltd.	0.181	Coal India Ltd.	0.218
10	Cipla Ltd.	Dr. Reddy's Laboratories Ltd.	0.234	Coal India Ltd.	0.182	Larsen & Toubro Ltd.	0.256
11	HDFC Bank Ltd.	Mahindra & Mahindra Ltd.	0.237	Mahindra & Mahindra Ltd.	0.188	HDFC Ltd.	0.287
12	Wipro Ltd.	Tata Motors Ltd.	0.242	Cipla Ltd.	0.188	Sun Pharmaceutical Inds. Ltd.	0.335
13	Larsen & Toubro Ltd.	Cipla Ltd.	0.262	Sun Pharmaceutical Inds. Ltd.	0.246	Vedanta Limited	0.436
14	Bharti Airtel Ltd.	Bajaj Auto Ltd.	0.272	NTPC Ltd.	0.271	Mahindra & Mahindra Ltd.	0.540
15	Tata Power Co. Ltd.	Bharat Heavy Electricals Ltd.	0.305	Bharti Airtel Ltd.	0.274	Hindalco Industries Ltd.	0.635

perts' opinions. This may significantly increase the development time and cost as the size of the rule base increases. These constraints are overcome significantly in this model as DS evidence theory is used here to generate automated fuzzy rule base. Inherent uncertainty in the design of the fuzzy rule base is also addressed at the same time.

Robustness of the proposed portfolio optimization model is another advantage. As performance of stocks are uncertain and dynamic in nature use of fuzzy return and and risk with fuzzy variance and skewness has considerably increased the robustness of the system.

All these improvements demonstrate the superiority of the proposed model over other existing expert systems.

Table 15. Empirical comparison of other ES with the proposed ES model

Article	Factors identified for the ES	Rule base size	Portfolio construction	Optimization Tool used	Results analysis and tested data source	Uncertainty handling tool
45	Different types of price and volume changes are used as parameters	108	Not Address	NA	Tested on real data from Warsaw Stock Exchange	Fuzzy sets and DS evidence theory
25	Criteria based on fundamental analysis technique for making rational investment decision within long term period	1406	Not Address	NA	firms whose equities are traded in Athens Stock Exchange leveraging from the expert opinions	No well-known tool is used
46	15 fundamental and 5 technical data are used as input to the ES	81	A model is constructed to maximize the total weighted ratings of the stocks incorporated in the recommended portfolio	Mixed integer linear programming model	Tested on real data from Istanbul Stock Exchange	Fuzzy sets
24	Seven factors are identified specified by experts mainly focusing on technical criteria	932	Based on different investment criteria provided by investors through questionnaires	No optimization tool is used	Tested on real data from Tehran Stock Exchange	Fuzzy Sets
This article	4 fundamental factors identified based on experts' consensus	81	A model is used to maximize the difference of fuzzy portfolio return and the risk free return to the weighted mean semi-variance of the assets	ACO	Tested on real data from Bombay Stock Exchange	Fuzzy Sets and DS evidence theory

11. Conclusion

In this research a novel fuzzy expert system model is designed and implemented to evaluate and rank the stocks under BSE. DS-evidence theory is used for the development of the fuzzy rule base to reduce the overall implementation time and cost of the system. A portfolio optimization model is constructed by considering the ratio of the difference of fuzzy portfolio return and risk free return to the weighted mean semi-variance of the assets as the objective function. ACO is used to obtain the optimum value of this portfolio.

As the outcome of this model found to be satisfactory, this can be implemented for any stock exchanges around the world but the selection of critical factors may vary over different stock exchanges. This fuzzy expert system model can be used to rank any set of alternatives based on the factors influencing them. The portfolio optimization model and the related ACO algorithm can be used to optimize any rank-preference based portfolios.

This work can be extended to include analysis of expert system on integrated data of assets from different stock exchanges and mutual funds. In this work ACO is used for optimization purpose. Researchers can use some other meta-heuristic algorithms such as simulated annealing, tabu search, particle swam optimization, genetic algorithm to find the portfolio.

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